

# Group Project Brief

## Admin

- There will be 10 groups in the class.
- One submissions for each section is required per group
- Not all members need to present at every presentation, but ideally everyone gets to present once (e.g. present at proposal but not at final presentation, or vice versa)

## Objective

This project aims to simulate real-world working conditions for a small team of Data Scientists to design, build and operationalise a machine learning use case. The successful outcome of this project for a group looks like something as follows:

- **Project Proposal / Request for Comment & Deck.** This is a technical proposal presentation to be done by Week 6 of the module and presented to the class to help the group align on their project design and approach. You may present more than 1 approach to get feedback on which to execute.
- **Project Code & Report.** The team then has to design and implement the code components of the project using industry standard open-source tools up to the point of deployment in the cloud, but these tools should be able to be deployed locally at the minimum in docker containers.
- **Project Final Presentation & Deck.** Groups are to present their final design architecture, show working artefacts demo (if appropriate) and present their final Tech Spec / ReadMe / Documentation of their pipeline and other MLOps / operational documentation and procedures.

A good project is one that applies the concepts taught in class thoughtfully in the deliverables. These concepts include but are not limited to (where applicable to the use-case):

| Topic                                      | Concepts                                                                                                                                                                  |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Introduction to software engineering tools | <ul style="list-style-type: none"><li>• Version control and Git</li><li>• Containers and docker</li></ul>                                                                 |
| Machine Learning Engineering and MLOps     | <ul style="list-style-type: none"><li>• Machine learning lifecycle</li><li>• Challenges of operationalizing ML</li><li>• Requirements for production ML systems</li></ul> |
| Data collection                            | <ul style="list-style-type: none"><li>• Cloud computing basic</li><li>• Data sources</li><li>• Data ingestion</li><li>• Storage</li><li>• Data labeling</li></ul>         |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data preparation       | <ul style="list-style-type: none"> <li>• Data cleaning, data splitting, feature engineering</li> <li>• Feature store</li> <li>• Preprocessing job using container</li> </ul>                                                                                                                                                                                                                                                        |
| Model development      | <ul style="list-style-type: none"> <li>• Model training and evaluation</li> <li>• Training job using container</li> <li>• Hyperparameter tuning</li> <li>• Distributed training</li> <li>• Compute infrastructure, GPU, and specialized chips</li> </ul>                                                                                                                                                                            |
| Model deployment       | <ul style="list-style-type: none"> <li>• Model registry</li> <li>• Deployment strategies: blue/green deployment, canary deployment, A/B testing, shadow deployment</li> <li>• Types of deployment: <ul style="list-style-type: none"> <li>◦ Batch inference</li> <li>◦ Online inference: real-time inference, asynchronous inference, serverless inference</li> </ul> </li> <li>• Inference pipelines</li> <li>• Scaling</li> </ul> |
| Workflow and pipelines | <ul style="list-style-type: none"> <li>• Automation and orchestration</li> <li>• Airflow - open-source platform for developing, scheduling, and monitoring batch-oriented workflows</li> </ul>                                                                                                                                                                                                                                      |
| Model monitor          | <ul style="list-style-type: none"> <li>• Importance of continuous monitoring</li> <li>• Monitoring model endpoints</li> <li>• Model monitor</li> <li>• Monitor data quality</li> <li>• Monitor model quality</li> <li>• Monitor bias drift</li> <li>• Monitor feature attribution drift</li> </ul>                                                                                                                                  |

## Use-case Topics

Groups can decide their own business use cases and find or create their own data synthetically. Please document all your assumptions you are making and clearly define your business / product and use-case, even if it is made up! Remember, more complex modeling or higher accuracy modeling does not necessarily mean higher grades! The aim of the project is to experience the engineering part of machine learning operationalisation; even poor-performing models but extremely rigorous MLOps implementation, monitoring and tech spec can be a A+ project! Focus on the engineering, solution architecting, ML operationalisation aspect of the project and you will do well!

## Deliverables (total 35 marks)

### Project Proposal / Request for Comment & Deck (7 marks)

- **Submit presentation deck in PDF.** (long deck doesn't mean more marks! In the real world, shorter decks that get the point across are more valued!)
- **5 mins presentation.**
- What is the business problem? What is the dataset? Who are the users and how are they going to use the machine learning system? What is the environment? How do you plan to implement? What is your high level design choices?

### Project Code & Report (14 marks)

- Code: 7 marks
  - **Functional code submitted as a zip file to elearn similar to labs and assignments**
  - Artefact / Environment completeness and tidiness
  - ReadMe (to execute code)
- Report: 7 marks
  - **PDF document max 8 pages, A4, font size 10 Arial**
  - Able to justify / defend design choices
  - Appropriate level of detail for deployment
  - Succinct to reflect industry practice

### Project Final Presentation & Deck (14 marks)

- Presentation Deck: 7 marks
  - **Submit presentation deck in PDF.** (long deck doesn't mean more marks! In the real world, shorter decks that get the point across are more valued!)
  - **10 mins presentation.**
  - Clarity, design, appropriate level of detail
- Communication skills: 7 marks
  - Graded at a group level (just like in real-life)
  - Cognitive flow of the presentation
  - Ability to communicate to targeted stakeholders
  - Able to justify / defend design choices